

Question ID: 54cb53cf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The number of zebras in a population in **2018** was **1.27** times the number of zebras in this population in **2014**. If the number of zebras in this population in **2014** is $p\%$ of the number of zebras in this population in **2018**, what is the value of p , to the nearest whole number?

Correct Answer: 79

Rationale

The correct answer is **79**. Let x represent the number of zebras in the population in **2014** and let y represent the number of zebras in the population in **2018**. It's given that the number of zebras in this population in **2018** was **1.27** times the number of zebras in this population in **2014**. It follows that the equation $y = 1.27x$ represents this situation. Dividing both sides of this equation by **1.27** yields $\frac{y}{1.27} = x$, or $(\frac{1}{1.27})y = x$. Therefore, the number of zebras in this population in **2014** is $\frac{1}{1.27}$ times the number of zebras in this population in **2018**. If the number of zebras in this population in **2014** is $p\%$ of the number of zebras in this population in **2018**, then $x = \frac{p}{100}y$. It follows that $\frac{1}{1.27} = \frac{p}{100}$, or $\frac{100}{1.27} = p$, which means p is approximately equal to **78.74**. Therefore, the value of p , to the nearest whole number, is **79**.